université de BORDEAUX

# Automatic tuning of the IO500 benchmark parameters with Bayesian Optimization

M2 Computer Science
Internship Report

Inria Centre at the University of Bordeaux

**[TODO:** student name**]**
Supervisors: **[TODO:** supervisor names**]**

August 13, 2026

# Contents

**1. Introduction** **1**
1.1. Context . . . . . 1
1.2. Problem statement . . . . . 1
1.3. Objectives of the internship . . . . . 1
1.4. Contributions . . . . . 1
1.5. Organisation of the report . . . . . 2

**Bibliography** **3**

# 1. Introduction

## 1.1. Context

- HPC applications are increasingly limited by I/O rather than by compute: data volumes grow faster than the storage system's ability to absorb them.
- Parallel file systems (Lustre, BeeGFS, GPFS) expose many tunable knobs (striping, number of storage targets, transfer size, process placement), and the right values are machine dependent.
- IO500 is the de-facto standard benchmark for ranking HPC storage systems: it combines bandwidth-oriented and metadata-oriented phases into a single score, and publishes a biannual list.
- Because IO500 is a *ranking*, sites have a direct incentive to tune it; but the published score of a system depends as much on how it was configured for the run as on the hardware itself.

## 1.2. Problem statement

- The IO500 submission rules require every write/create phase to run for at least 300 s (stonewall timer set to 300 s); a run whose write phase finishes earlier is marked **invalid**.
- Consequence: one *valid* IO500 measurement costs roughly 30 min of exclusive cluster time.
- The parameter space explored in this work has **[TODO:** confirm final number**]** $\approx$ 7,840 configurations; measuring all of them validly with 3 repetitions would take $\approx$ 12 000 h of wall-clock time, i.e. more than one year on the target machine.
- Exhaustive tuning is therefore out of reach, and practitioners fall back on expert intuition or on greedy, one-parameter-at-a-time sweeps.
- **Research question:** can a smarter search reach a near-optimal IO500 configuration in far fewer evaluations, and which search method actually does so?

## 1.3. Objectives of the internship

- Build a *cheap* but faithful ground truth of the IO500 response surface on a real cluster.
- Evaluate Bayesian Optimization (BO) as a tuning strategy for IO500, and determine *which* BO configuration works, since "Bayesian Optimization" is not a single method but a family (surrogate model $\times$ acquisition function $\times$ exploration parameter $\times$ budget).
- Compare BO against random search, exhaustive search, and a human specialist.
- Package the resulting methodology and configurations inside the IOPS framework so that the study is reproducible by third parties.

## 1.4. Contributions

- **A cheap ground-truth methodology.** Running the sweep with a shortened stonewall (10 s) and a small dataset produces *invalid* IO500 scores that nevertheless preserve the relative ranking of configurations. The full grid was measured in $\approx$ 183 h instead of more than a year.

- **A reusable cache.** Every `parameters` → `SCORE` pair is stored in an SQLite cache (`io500.db`), so any search strategy can be replayed offline at zero cluster cost with `iops run -cache-only`.
- **A systematic comparison of BO setups**: **[TODO:** confirm] 83 BO configurations (surrogate ∈ {GP, RF, ET}, acquisition ∈ {EI, PI, LCB}, exploration $\xi/\kappa/n_{\text{init}}$, budget ∈ $\{10, 15, 20, 30, 40\}$), each repeated over 10 seeds.
- **A sensitivity study** isolating which BO knob actually drives quality.
- **A validation campaign** re-running the best configurations with a *valid* IO500 setup on the cluster, to check that the cheap proxy really does preserve the ranking.
- **A human baseline**: a specialist challenge in which an expert submits their best guesses through a form, run under exactly the same conditions as the automated search.

### 1.5. Organisation of the report

- **Context** presents the host laboratory, the team and the internship framework.
- **Background** introduces IO500, parallel file system tuning and Bayesian Optimization.
- **Related work** reviews existing approaches to I/O auto-tuning.
- **Tools and experimental setup** describes the tools, the target platform and the complete installation procedure.
- **Methodology** details how the ground truth was built and how the search strategies were compared.
- **Results** reports and analyses the measurements.
- **Discussion** covers the limitations and threats to validity, and **Conclusion** summarises the work and its perspectives.

# Bibliography

## References


[1] IO500 Foundation. The IO500 list. `https://io500.org/`, 2026. Accessed: August 13, 2026.

[2] IO500 Foundation. IO500 submission rules. `https://io500.org/rules/submission`, 2026. Accessed: August 13, 2026.

[3] TADaaM, Inria. IOPS: Integrated orchestration for parametric studies. `https://gitlab.inria.fr/hpc_io/iops`, 2026. Accessed: August 13, 2026.

[4] The scikit-optimize contributors. scikit-optimize: Sequential model-based optimization in Python. `https://scikit-optimize.github.io/`, 2026. Version TO-COMPLETE.

[5] TO-COMPLETE. To-complete: 2022 I/O performance study. `https://inria.hal.science/hal-03753813/`, 2022.

[6] TO-COMPLETE. To-complete: IO500 tuning on CLAIX23. In *IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops)*, 2025.

[7] TO-COMPLETE. To-complete: IO500 benchmark reference paper, 2026.